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RETURN STATEMENT IN PYTHON

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1. WHAT IS A RETURN STATEMENT?

The return statement is used inside a function to:

- Send a value back to the place where the function was called
- Stop the execution of the function

Once a return statement is executed, the function ends immediately.

2. WHY DO WE NEED A RETURN STATEMENT?

We use return when:

- We want to get a result from a function
- We want to reuse the output in another part of the program
- We want the function to perform calculation and give back the result

A function without return only performs an action,
but a function with return produces a result.

3. FUNCTION WITHOUT RETURN STATEMENT

Example:

```
def add(a, b):  
    print(a + b)
```

```
add(10, 5)
```

Output:

15

Explanation:

- The function prints the result
- The value cannot be stored or reused
- The function returns None by default

4. FUNCTION WITH RETURN STATEMENT

Example:

```
def add(a, b):  
    return a + b
```

```
result = add(10, 5)  
print(result)
```

Output:
15

Explanation:

- The function returns the value
- The returned value is stored in a variable
- The result can be reused later

5. REAL WORLD EXAMPLE (EASY TO UNDERSTAND)

Think of a vending machine:

- You insert money (input)
- The machine processes it
- It gives you a snack (output)

The snack is like the RETURN value.

If the machine only showed the snack but did not give it to you, you cannot use it.

Return = giving the snack

Print = only showing the snack

6. DIFFERENCE BETWEEN RETURN AND PRINT

PRINT:

- Displays output on the screen
- Cannot be reused
- Used for debugging or display purpose only

RETURN:

- Sends value back to the caller

- Can be stored and reused
- Used for actual program logic

Example:

```
def square(n):  
    print(n * n)
```

```
x = square(4)  
print(x)
```

Output:

16

None

Because print does not return a value.

7. FUNCTION RETURNING MULTIPLE VALUES

A function can return more than one value using commas.

Example:

```
def calculate(a, b):  
    return a + b, a - b, a * b
```

```
sum_, diff, prod = calculate(10, 5)  
print(sum_, diff, prod)
```

Explanation:

- Python returns values as a tuple
- Multiple results can be returned at once

8. MORE THAN ONE RETURN STATEMENT IN A FUNCTION

A function can have multiple return statements,
but only ONE return will execute.

Example:

```
def check_even(num):
```

```
if num % 2 == 0:  
    return "Even"  
return "Odd"
```

```
print(check_even(7))
```

Explanation:

- If condition is true, first return runs
- Otherwise, second return runs
- Function exits as soon as return executes

9. IMPORTANT POINTS TO REMEMBER

- Code after return will never execute
- A function without return returns None
- Return can return any data type:
int, float, string, list, tuple, dictionary, object
- Return is used when we want output for further use

10. EXAM-FRIENDLY DEFINITION

"The return statement is used to terminate a function and send a value back to the calling function."

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END OF NOTES

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